

pyCHM derivations

First-principles group theory of the Minimal and Next-to-Minimal Composite Higgs

Generated for the pyCHM library

June 21, 2026

Abstract

This note derives, from group theory alone, every quantity that `pyCHM` treats as *derived* (as opposed to model input, or numerically anchored to the independent `pypngb` engine). Each step is tied to the function that implements it and the test that verifies it in closed form, so the document and the code are one object: if a derivation here is wrong, a named test fails. The reference physics is Murnane, *The Landscape of Composite Higgs Models* (arXiv:2606.18364); conventions match the library exactly. Nothing here assumes the thesis is correct — the closed forms are recomputed from the generators, and where the thesis is the only source (the App. A7 form factors) it is cross-checked against the `pypngb`-anchored eigenvalue route (§5).

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1 The CCWZ framework

A composite Higgs is a set of pseudo-Nambu–Goldstone bosons (pNGBs) of a spontaneously broken global symmetry $G \rightarrow H$. Write the generators of $\mathfrak{g} = \text{Lie}(G)$ as the unbroken $T^a \in \mathfrak{h}$ and the broken $X^{\hat{a}} \in \mathfrak{g}/\mathfrak{h}$. The pNGBs $\pi^{\hat{a}}(x)$ parameterise the coset through the Goldstone matrix

$$U(\pi) = \exp\left(\frac{i}{f} \pi^{\hat{a}} X^{\hat{a}}\right), \quad (1)$$

with f the decay constant. Fermions are embedded in (generally incomplete) representations R of G ; the Higgs dependence of every elementary–composite mixing is the matrix element of U in R ,

$$\langle c | U_R(\pi) | E \rangle, \quad (2)$$

between the embedding E of the elementary field and a composite state c . The central claim of the library is that *these matrix elements are the only source of the s_h -dependence*, and that they are computable in closed form from the generators — never fitted. The rest of this document carries out that computation for $\text{SO}(5)/\text{SO}(4)$ (MCHM) and $\text{SO}(6)/\text{SO}(5)$ (NMCHM).

2 Minimal Composite Higgs: $\text{SO}(5)/\text{SO}(4)$

2.1 Generators and conventions

$\text{SO}(5)$ acts on indices $0, \dots, 4$. The unbroken $\text{SO}(4)$ acts on $0, \dots, 3$; the coset direction is index 4. In the vector (the **5**) the Hermitian generators are

$$(T^{AB})_{CD} = -i(\delta_{AC}\delta_{BD} - \delta_{AD}\delta_{BC}), \quad 0 \leq A < B \leq 4. \quad (3)$$

The six T^{ab} ($a < b \leq 3$) generate $\text{SO}(4)$; the four broken generators are $X^{\hat{a}} = T^{\hat{a}4}$. With the Higgs vacuum expectation value along the $\hat{a} = 3$ direction, the only field that survives in unitary gauge is the physical Higgs h , and U becomes a single planar rotation.

► **Implemented in:** `groups.so5.gen_vector`, `groups/so5.py` **Verified by:** `test_so4_unbroken_generators_close_t`

2.2 The vector Goldstone matrix

Exponentiating the broken generator $X^3 = T^{34}$ with angle $\theta = h/f$ gives a rotation in the $(3, 4)$ plane,

$$U_5(\theta) = \exp(i\theta T^{34}) = \begin{pmatrix} \mathbb{K}_3 & & \\ & c_h & s_h \\ & -s_h & c_h \end{pmatrix}, \quad s_h \equiv \sin \theta, \quad c_h \equiv \cos \theta = \sqrt{1 - s_h^2}. \quad (4)$$

The library stores everything in the variable $s_h = \sin(h/f)$ with $c_h = \sqrt{1 - s_h^2}$, avoiding an arcsin round-trip (machine precision). Eq. (4) is the thesis “explicit Goldstone matrix.”

► **Implemented in:** `groups.so5.U_vector`, `so5.U_vector_sym` **Verified by:** `test_U_vector_matches_thesis_goldst`

2.3 Lifting to tensor representations

A rank- k $\text{SO}(5)$ tensor irrep is carried by the totally symmetric-traceless or totally antisymmetric part of $\mathbf{5}^{\otimes k}$. The Goldstone acts on each index,

$$(U_R)_{ba} = \langle E_b | U_{\mathbf{5}^{\otimes k}} E_a \rangle = \text{Tr}(E_b^\dagger U E_a U^\top) \quad (\text{rank } 2), \quad (5)$$

in an orthonormal basis $\{E_a\}$ of the irrep. The first three rungs are the **5** (rank 1), the **10** (antisymmetric rank 2) and the **14** (symmetric-traceless rank 2). The construction is dimension-agnostic, so the same code produces the **30** (rank-3 symmetric) and, with $n = 6$, the $\text{SO}(6)$ tower of §3.

► **Implemented in:** `tensors.tensor_basis`, `tensors.U_rep_tensor`, `groups.so5.U_rep` **Verified by:** `test_tensors.py` (dimension, unitarity, homomorphism)

2.4 $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ branching

The unbroken $\text{SO}(4)$ on indices $0, \dots, 3$ is $\text{SU}(2)_L \times \text{SU}(2)_R$ via the 't Hooft combinations

$$J_i^{L/R} = \frac{1}{2} \left(\frac{1}{2} \epsilon_{ijk} T^{jk} \pm T^{0i} \right), \quad i = 1, 2, 3. \quad (6)$$

Diagonalising the two Casimirs $C_L = \sum_i (J_i^L)^2$, $C_R = \sum_i (J_i^R)^2$ on the irrep basis (they commute) labels each sub-multiplet by (j_L, j_R) with $C = j(j+1)$. The branchings are

$$\mathbf{5} = \left(\frac{1}{2}, \frac{1}{2} \right) \oplus (0, 0), \quad (7)$$

$$\mathbf{10} = \left(\frac{1}{2}, \frac{1}{2} \right) \oplus (1, 0) \oplus (0, 1), \quad (8)$$

$$\mathbf{14} = (1, 1) \oplus \left(\frac{1}{2}, \frac{1}{2} \right) \oplus (0, 0). \quad (9)$$

► **Implemented in:** `decompose.so4_decompose`, `decompose._su2_casimirs` **Verified by:** `test_branchings_match_th`

2.5 Dressings are matrix elements (derive, don't fit)

A claimed closed form $f(\theta)$ for an elementary-composite mixing is admissible only if it lies in the span of the single-basis overlaps $\{\langle E_b | U_R | E \rangle\}$. The library solves for the exact minimal-norm composite tensor c with $\langle c | U_R | E \rangle = f$ and *raises* if f is not realisable. This is the from-scratch guarantee: every dressing factor in the model files is *proven* to be a Goldstone matrix element, with symbolic residual exactly 0.

► **Implemented in:** `groups/derive.py::solve_composite` **Verified by:** `test_dressings_are_matrix_elements`

2.6 The 5-5-5 coefficients

Embed the up-component of q_L in the bidoublet, $E_{q_L} = (e_0 + e_3)/\sqrt{2}$, and t_R in the $\text{SO}(4)$ singlet $E_{t_R} = e_4$. From (4), $Ue_4 = s_h e_3 + c_h e_4$ and $Ue_3 = c_h e_3 - s_h e_4$, $Ue_0 = e_0$, so

$$\langle q_L | U_{\mathbf{5}} | t_R \rangle = \left\langle \frac{e_0 + e_3}{\sqrt{2}} \mid s_h e_3 + c_h e_4 \right\rangle = \frac{s_h}{\sqrt{2}}, \quad \left| \langle q_L | U | t_R \rangle \right|^2 = \frac{s_h^2}{2}, \quad (10)$$

$$\langle q_L | U_{\mathbf{5}} | q_L \rangle = \frac{1}{2} (1 + c_h) = \cos^2 \frac{h}{2f}. \quad (11)$$

These are exactly the $\Pi_L \sim s_h^2/2$, $\cos^2(h/2f)$ structures of the thesis 5-5-5 form factors, here *derived* from the overlaps.

► **Implemented in:** `so5.overlap_sym`, `mchm5.formfactor_pieces` **Verified by:** `test_coeffs_5_5_5`

2.7 The 14 channel weights and the 4/5 enhancement

The t_R embeds in the $\text{SO}(4)$ singlet of the **14**, the unit-normalised symmetric-traceless tensor

$$S = \frac{1}{\sqrt{20}} \text{diag}(1, 1, 1, 1, -4), \quad \text{Tr} S^2 = \frac{4+16}{20} = 1. \quad (12)$$

Dress it, $S \mapsto USU^\top$, and project onto the $\text{SO}(4)$ channels of the **14**. The squared lengths in the singlet $(1, 1)$, bidoublet $(\frac{1}{2}, \frac{1}{2})$, and $(1, 1) \rightarrow (3, 3)$ channels are the closed trigonometric *channel weights*

$$W_{11} = \frac{(4c_h^2 - s_h^2)^2}{16}, \quad W_{22} = \frac{5}{2} s_h^2 c_h^2, \quad W_{33} = \frac{15}{16} s_h^4, \quad (13)$$

which satisfy unitarity exactly:

$$W_{11} + W_{22} + W_{33} = \left(1 - \frac{5}{2} s_h^2 + \frac{25}{16} s_h^4\right) + \frac{5}{2} s_h^2 (1 - s_h^2) + \frac{15}{16} s_h^4 = 1. \quad (14)$$

The thesis writes the t_R self-energy with an overall coupling $Y_T \sqrt{4/5}$. That normalisation is *not* an external input: it is the squared index-4 component of the embedding itself,

$$\boxed{|S_{44}|^2 = \left(\frac{-4}{\sqrt{20}}\right)^2 = \frac{16}{20} = \frac{4}{5}} \quad \Longleftrightarrow \quad \sqrt{\frac{4}{5}} = |S_{44}|. \quad (15)$$

Index 4 is the coset-singlet direction through which the top mass is generated; the **5**-singlet has weight 1 there (the unenhanced reference), so the 4/5 is a genuine representation-dependent enhancement. The thesis singlet-channel prefactor then follows as a *prediction*,

$$\frac{(4c_h^2 - s_h^2)^2}{20} = |S_{44}|^2 W_{11} = \frac{4}{5} \cdot \frac{(4c_h^2 - s_h^2)^2}{16}, \quad (16)$$

which would fail for any inconsistent thesis number — the check is no longer a tautology that reads 4/5 off the prefactor 1/20.

► **Implemented in:** `decompose.channel_weights_sym`, `so5.embedding_sym` **Verified by:** `test_14_channel_weights`, `test_4_5_is_derived_from_the_embedding`

3 Next-to-Minimal Composite Higgs: $\text{SO}(6)/\text{SO}(5)$

3.1 Coset and broken generators

$\text{SO}(6)$ acts on indices $0, \dots, 5$. The unbroken $\text{SO}(5)$ acts on $0, \dots, 4$; the coset direction is index 5. The five broken generators split, under the $\text{SO}(4) \subset \text{SO}(5)$, into the Higgs bidoublet and a singlet:

$$X_B^a = T^{a5} \quad (a = 0, 1, 2, 3) \quad (\text{the doublet } h_1, \dots, h_4), \quad X_S = T^{45} \quad (\text{the singlet } s). \quad (17)$$

This is the thesis decomposition (its 6th index is our index 5). The 15 generators close the $\mathfrak{so}(6)$ algebra; the ten T^{ab} with $a < b \leq 4$ are the unbroken $\text{SO}(5)$.

► **Implemented in:** `so6.gen_vector`, `so6.UNBROKEN/BROKEN` **Verified by:** `test_so6_generators_close_the_algebra`, `test_coset_split_10_plus_5`

3.2 The five-pNGB Goldstone matrix (closed form)

In unitary gauge the Higgs points along $\hat{a} = 3$ and the singlet along X_S , so the vector Goldstone is generated by

$$G = \theta_h L^{(3,5)} + \theta_s L^{(4,5)}, \quad \theta_h = \frac{h}{f}, \quad \theta_s = \frac{s}{f}, \quad (18)$$

with $L^{(ab)}$ the real antisymmetric rotation generators. G is supported on the three indices $\{3, 4, 5\}$ and has rank 2, hence satisfies $G^3 = -\Theta^2 G$ with $\Theta = \sqrt{\theta_h^2 + \theta_s^2}$. Rodrigues' formula then gives the Goldstone in closed form,

$$U_6(\theta_h, \theta_s) = \mathbb{1} + \frac{\sin \Theta}{\Theta} G + \frac{1 - \cos \Theta}{\Theta^2} G^2. \quad (19)$$

Acting on the vacuum e_5 reproduces the thesis Goldstone field (eq. 474 there):

$$\Phi = U_6 e_5 = \left(0, 0, 0, \theta_h \frac{\sin \Theta}{\Theta}, \theta_s \frac{\sin \Theta}{\Theta}, \cos \Theta\right)^\top, \quad (20)$$

i.e. $\Phi = \frac{1}{\Theta} \sin \Theta (h_1, \dots, h_4, s, \Theta \cot \Theta)$ with $\Theta = \varphi/f$, $\varphi = \sqrt{\sum h_i^2 + s^2}$. Setting $\theta_s = 0$ returns the MCHM planar rotation (4) (with coset index $4 \rightarrow 5$).

► **Implemented in:** `so6.U6_vector`, `so6.U6_vector_sym`, `so6.goldstone_vacuum` **Verified by:** `test_goldstone_vacuum`, `test_U6_vector_orthogonal_identity_homomorphism`

3.3 The representation tower

The fermion-partner irreps of $\text{SO}(6) \cong \text{SU}(4)$ are built and dressed by the same machinery: tensors via $n = 6$, spinors via the Clifford algebra. Their dimensions, $\text{SO}(5)$ branchings (from the quadratic Casimir $C_5 = \sum_{a < b \leq 4} (T^{ab})^2$), and $\text{SO}(4)$ content are

irrep	construction	dim	$\text{SO}(5)$ branching
6	vector	6	5 $\oplus$ 1
15	antisym. rank 2 (adjoint)	15	10 $\oplus$ 5
20'	sym.-traceless rank 2	20	14 $\oplus$ 5 $\oplus$ 1
10, $\overline{10}$	(anti)self-dual 3-form	10	10 (each)
4, $\overline{4}$	Weyl spinor	4	4

The Casimir calibration $C_5 \in \{0, \frac{5}{2}, 4, 6, 10\} \leftrightarrow \{\mathbf{1}, \mathbf{4}, \mathbf{5}, \mathbf{10}, \mathbf{14}\}$ identifies each sub-irrep with its multiplicity; the antisymmetric 3-form is 20-dimensional and splits, by the $\pm i$ eigenspaces of the Hodge dual $(\star T)_{ijk} = \frac{1}{3!} \epsilon_{ijklmn} T_{lmn}$, into the self-dual **10** and anti-self-dual **$\overline{10}$** (the $\text{SU}(4)$ **10** $+$ **$\overline{10}$**). The spinors come from six 8×8 Euclidean gammas $\Gamma^A = (\sigma_1 \otimes \gamma_{\text{SO}(5)}^a, \sigma_2 \otimes \mathbb{1})$, split by chirality $\chi = \sigma_3 \otimes \mathbb{1}$; the **4** carries no $(\frac{1}{2}, \frac{1}{2})$ bidoublet, which is exactly the thesis reason the NM4DCHM uses the **6** (not the **4**) for q_L .

► **Implemented in:** `so6.rep_basis`, `so6.U6_rep`, `so6.so5_content`, `so6_spinors.py` **Verified by:** `test_so6.py`, `test_so6_spinors.py`

3.4 Two-field dressing and the singlet pNGB

The **6** Goldstone dresses the elementary–composite overlaps with *both* angles. With $E_{q_L} = (e_0 + e_3)/\sqrt{2}$ and $E_{t_R} = e_5$, no embedding has support on index 4, so every overlap depends on θ_h, θ_s only through quantities even in θ_s ; the potential is even in s , the vacuum sits at $\langle s \rangle = 0$, and the singlet mass is the s -curvature of the Coleman–Weinberg potential there. Because the q_L bidoublet dressing carries explicit θ_h -dependence distinct from the radial Θ the t_R feels, $V(h, s)$ is genuinely two-dimensional and the singlet acquires a finite, calculable mass — the defining NMCHM observable, absent in the MCHM. At $\theta_s = 0$ the mass matrices reduce *entry-for-entry* to the `pypngb`-anchored 5-5-5 model.

► **Implemented in:** `nmchm6.py` (`spec`, `assemble`, `potential`, `singlet_mass2`) **Verified by:** `test_nmchm6.py` (`reduction`, `even-in-s`, `singlet mass`)

4 The Coleman–Weinberg potential and observables

The one-loop pNGB potential is the supertraced kernel

$$V(s_h) = \sum_i \frac{c_i}{64\pi^2} m_i^2(s_h)^2 \log m_i^2(s_h), \quad c_i = \{+3, +6, -12\} \quad (21)$$

for {neutral vector, charged vector, coloured Dirac}, with $m_i(s_h)$ the eigenvalues of the s_h -dependent mass matrices. Writing $V = -\gamma s_h^2 + \beta s_h^4$, the vacuum, Higgs mass and electroweak scale are the closed forms

$$\xi = \sin^2 \frac{\langle h \rangle}{f} = \frac{\gamma}{2\beta}, \quad m_h^2 = \frac{8\beta}{f^2} \xi(1 - \xi), \quad v_{\text{EW}} = f\sqrt{\xi}, \quad (22)$$

and the Barbieri–Giudice tuning is $\Delta_{\text{BG}} = \max_i |\partial \ln f / \partial \ln x_i|$ over the fundamental Lagrangian masses x_i . These relations are validated symbolically; the full mass matrices and the resulting $\xi, m_t, m_h, \Delta_{\text{BG}}$ are anchored numerically to the independent `pypngb` engine to $< 0.1\%$.

► **Implemented in:** `routes.py`, `potential.py`, `spectrum.py`, `tuning.py` **Verified by:** `test_cw_kernel_and_coefficients.py`, `test_higgs_mass_scaling_closed_form`, `test_anchors.py`

5 The form-factor route and the oracle boundary

The thesis App. A7 form factors A_L, A_R, A_M, B are transcribed in `mchm5`; transcription proves faithful copying, not physical correctness. They are confronted with the `pypngb`-anchored eigenvalue route by comparing the top mass two ways. With the *custodial* parametrisation the form-factor route requires (a single q_L coupling L_q , i.e. $\Delta_{uL} = \Delta_{dL}$), the two coincide as $s_h \rightarrow 0$,

$$\frac{m_t^{\text{ff}}(s_h)}{m_t^{\text{eig}}(s_h)} = 1 - 0.755 s_h^2 + \mathcal{O}(s_h^4) \xrightarrow{s_h \rightarrow 0} 0.9999997, \quad (23)$$

the finite- s_h gap being the form-factor route’s leading-order-in- s_h^2 truncation of the exact (non-polynomial) eigenvalue mass. The App. A7 form factors are therefore the correct leading-order two-point functions of the `pypngb`-anchored mass matrix — validated, not merely transcribed.

► **Implemented in:** `mchm5.formfactor_pieces`, `mchm5.fermion_mass` **Verified by:** `test_formfactor_route.py`

6 Beyond SO(N): the coset abstraction and SU(4)/Sp(4)

Everything above is built from one group-agnostic engine (`pychm.groups`): a coset G/H is a list of unbroken generators of H and broken generators X^a in some representation, and the Goldstone is $U(\pi) = \exp(i\pi^a X^a)$ – in unitary gauge a rank- ≤ 2 rotation, so the closed Rodrigues form (19) serves every case. The MCHM and NMCHM are the instances $\text{SO}(5)/\text{SO}(4)$ and $\text{SO}(6)/\text{SO}(5)$; the *same code* builds $\text{SU}(N)/\text{Sp}(N)$ cosets, with the $\text{SU}(N)$ Gell-Mann generators and the $\text{USp}(N)$ subalgebra obtained from the Cartan involution $\theta(X) = \Omega \bar{X} \Omega^{-1}$ (Ω the symplectic form): `usp(N)` is the $\theta = -1$ eigenspace, the coset the $\theta = +1$ eigenspace.

SU(4)/Sp(4) $\cong$ SO(6)/SO(5). The minimal pseudoreal Ferretti–Sannino coset $\text{SU}(4)/\text{Sp}(4)$ has $15-10 = 5$ broken generators — the same five pNGBs (Higgs doublet + singlet) as $\text{SO}(6)/\text{SO}(5)$. Under $\text{SU}(4) \cong \text{Spin}(6)$, $\text{Sp}(4) \cong \text{Spin}(5)$ the representations map as

$$\mathbf{4}_{\text{SU}(4)} = \mathbf{4}_{\text{spinor}}^{\text{SO}(6)}, \quad \mathbf{6}_{\text{SU}(4)}^{[\text{antisym}]} = \mathbf{6}_{\text{vector}}^{\text{SO}(6)}, \quad \mathbf{15}_{\text{SU}(4)} = \mathbf{15}^{\text{SO}(6)}. \quad (24)$$

The library builds $\text{SU}(4)/\text{Sp}(4)$ independently (its own generators, Goldstone, and the antisymmetric $\mathbf{6} = [\mathbf{4} \otimes \mathbf{4}]_A$) and cross-checks it against the existing $\text{SO}(6)$ code, which is an *exact oracle*: the $\mathbf{4}$ branches to a single $\mathbf{4}$ of $\text{Sp}(4)$ and the $\mathbf{6}$ to $\mathbf{1} \oplus \mathbf{5}$, matching the $\text{SO}(6)$ spinor and vector; and the strong statement of the isomorphism — the $\mathbf{6}$ generators span the full 15-dimensional $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ algebra and act irreducibly (commutant = scalars, by Schur) — holds, so the $\text{SU}(4)$ antisymmetric $\mathbf{6}$ *is* the irreducible $\text{SO}(6)$ vector. The new coset thus reproduces the NMCHM group theory while sharing all of its tooling.

► **Implemented in:** `groups/lie.py` (`su_generators`, `sp_subalgebra`), `groups/coset.py`, `groups/su4sp4.py`
Verified by: `test_su4sp4.py`

Provenance: derived vs. input vs. anchored

Derived (this document)

the Goldstone matrices U, U_6 ; the rep lifts; the $\text{SO}(4)/\text{SO}(5)$ branchings; the 5-5-5 coefficients; the $\mathbf{14}$ channel weights and the $4/5 = |S_{44}|^2$ enhancement; the full $\text{SO}(6)$ tower. Each closes symbolically (a named test asserts `simplify(·) = 0`).

Input (model structure)

the partner content — which composite multiplets exist, the embeddings, the partner masses. Taken from the thesis/`pypngb`, not derived.

Anchored

the full mass matrices and the physical observables, validated against `pypngb` ($< 0.1\%$). The NMCHM has no public engine and is anchored indirectly by its exact reduction to the 5-5-5 at $\langle s \rangle = 0$, with the new singlet observable cross-checked internally (stencil independence + parabolic fit).

After §2.7 and §5, no quantity that feeds an observable is a thesis read-off: correctness rests on `pypngb` + group theory.